

(TR-104) PYTHON WITH AI/ML

Training Day 23 Report

Date: 11 February 2026

Topic Covered: Classical NLP – Text Classification, Model Evaluation & NLP Pipeline Engineering

The twenty-third day of training focused on implementing classical NLP concepts including text classification models, evaluation techniques, and understanding limitations of traditional NLP systems. The day also covered building an end-to-end NLP pipeline and analyzing model performance.

- Converted text data into Bag of Words (BoW) and TF-IDF representations.
- Built a Spam Classifier using Naive Bayes algorithm.
- Implemented Logistic Regression model for spam classification.
- Compared model performance and analyzed accuracy differences.
- Performed error analysis by identifying false positives and false negatives.
- Understood how models use vectorized features for prediction.
- Learned importance of training vs testing phase in machine learning models.
- Studied evaluation metrics including accuracy, precision, recall, and F1-score.
- Understood overfitting and importance of generalization in models.
- Learned train-test split and prevention of data leakage.
- Built an end-to-end NLP pipeline: preprocessing, vectorization, training, prediction, evaluation.
- Studied limitations of classical NLP such as context understanding, synonym issues, and word order problems.
- Understood why classical NLP is insufficient for semantic understanding of language.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/NLP>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/vectorization-in-nlp-0ccc759ded32>